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Changes in Propeller Performance Due to Ground Proximity

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We examine extreme ground effect for a range of off-of-the-shelf propellers suitable for small UAVs, with the ground-plate behind and alternatively ahead of the propeller. Propeller size is aimed to be above that plagued by low-Reynolds number effects, but still realizable with conventional electric hobby-type drive and control systems. Parameters include propeller diameter, solidity, and pitch, in various combinations, to elucidate trends in thrust-enhancement and/or power-reduction with closer ground-proximity. Generally, the lower the ratio of pitch to diameter, the stronger the ground-effect, with symmetric result whether the ground-plane is behind or ahead of the propeller. Power-required at constant thrust is used as a metric for ground-effect, in lieu of figure of merit. Comparing with classical theory and other experimental results, we propose a phenomenological expression combining the aforementioned parameters, to reduce the various parameter studies approximately to one curve of power-reduction in ground effect, to the normalized distance from the ground.

I. Nomenclature

A	=	Propeller Disk area (m^2)
C_T	=	Propeller Thrust Coefficient; $C_T = \frac{T}{\rho n^2 D^4}$
C_P	=	Propeller Power Coefficient; $C_P = 2\pi\, C_Q = \frac{P}{\rho n^3 D^5}$
C_Q	=	Propeller Torque Coefficient; $C_Q = \frac{Q}{\rho n^2 D^5}$
D	=	Propeller Diameter, (m)
H	=	Distance to the ground, (m)
n	=	Propeller RPS
P	=	Power, (W)
γ	=	Propeller Pitch, $in.$
P_c	=	Corrected Power (W)

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P_{OGE} = Propeller out of Ground Effect (OGE) (W)

Q = Torque (Nm)

S = Propeller Blade Projected Area (m^2)

T = Thrust, (N)

ρ = Air Density, (kg/m^3)

σ = Propeller Solidity; $\sigma = \frac{S}{0.25\pi D^2}$

1. Introduction

Ground effect on lifting rotor/propeller performance has been studied theoretically [1] and experimentally ([2], [4]-[6], [8], 14]) for decades. The general idea is that placing the propulsor adjacent and parallel to a ground-plane offers two benefits: an increase in thrust-produced, and a decrease in power-required, both for a given rotational speed. Historically, most investigations focus on helicopter rotors, which rarely have a spanwise twist. Here we follow the widespread burgeoning of small-scale propulsor experiments, made possible by ever-improving components from electric radio-control "hobby" aviation, to consider parameter variations not broadly available in the literature. In particular, we consider the extreme ground effect, where the ratio of ground plane stand-off distance to propeller diameter is 0.1 or less. We reverse the propeller, to examine as it were positive and negative ground effect. And we attempt to resolve real or perceived inconsistencies in the literature, owing to variations in propeller pitch and solidity, which sometimes confound the arrival at a clear trend.

Propeller ground-effect theory dates back at least Betz's model, in 1937 [1]. One way of presenting results is the normalized power ratio P/P_{OGE} vs. h/D. h/D is the ratio of the distance from the propeller reference-point above the ground, h, to the propeller diameter, D. The in-ground-effect (IGE) thrust at any given rpm will be higher than out-of-ground-effect (OGE; h/D --> ∞), and correspondingly, the power-required will also be lower. As h/D is reduced, the throttle setting (that is, rpm) can be reduced, to keep thrust constant. At every such setting, at every h/D, there will be a corresponding IGE value of Power (P). This is plotted as P/P_{OGE} for Betz's theory, together with a smattering of experimental results (Zbrozek [2], Gilad, Chopra and Rand [3], Knight and Heffner [5], and Tanner and Overmeyer [7]), in Figure 1. Zbrozek studied rotor blades with no twist and constant pitch at 150RPM. Gilad studied a rotor of 1.37m diameter with no twist at 70 RPM. Fradenburgh studied a rotor of 0.61m diameter with no twist at 5700 RPM. Tanner and Overmeyer tested a rotor of 3.38m at 1150RPM. There is a decreasing trend for P/P_{OGE} as h/D decreases, with 0.3 h/D as something of a crossover point (noted as a dashed red line in Figure 1). Indeed, Betz [1] calls 0.3 > h/D as "small", and 0.3 < h/D as "large", with different theoretical approximations for each. But the experimental results in Figure 1, appear to pass through h/D = 0.3 smoothly. Results from [2], [3] and [5] nearly overlap, with [7] the outlier and all predict smoother drop-off of IGE power-required vs. h/D, than Betz's theory. And for all results except for Tanner and Overmeyer [7], h/D = 1 is essentially infinity.

An alternative presentation is to plot the ratio of IGE to OGE thrust, for a given rpm, vs. h/D (Figure 2). Experimental data from Fradenburgh [4] (Sikorsky S-60, 22m rotor, 4 blades, relatively "high" solidity) and Law [8] (Bell-47 J-2, 11.3m rotor, 2 blades, relatively "low" solidity) are shown. Difference between the two is considerable and suggests that a geometric or aerodynamic variable besides just h/D is material. Candidates include pitch and solidity.

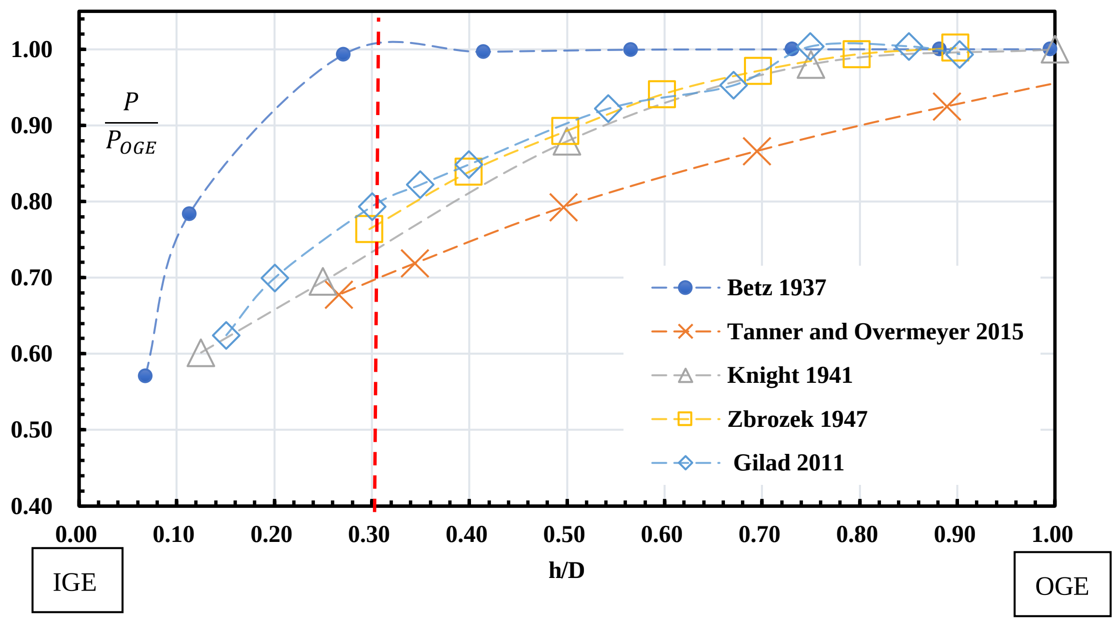

Figure 1. Ratio of in-ground-effect (IGE) to out-of-ground- effect (OGE) power-required at constant thrust, vs. h/D

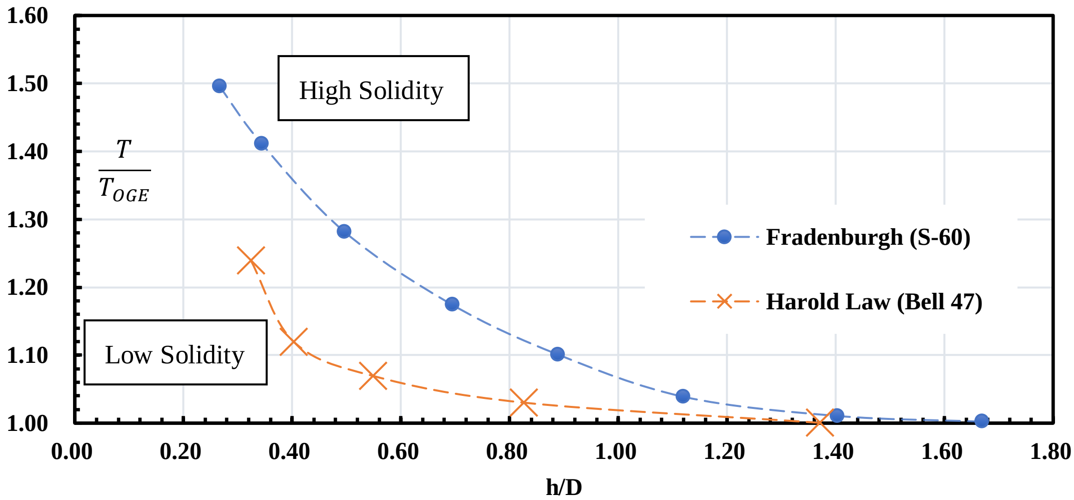

Figure 2. Experimental data on IGE to OGE thrust ratio, vs. h/D.

In the experiments of Knight and Hefner [5] (1.52m rotor, no twist and various pitch angles from 4 degrees to 14 degrees at 900 RPM; Figure 3), IGE power is seen to be affected by blade pitch angle (we will subsequently use the terminology more common for propellers: "pitch", in linear displacement per revolution, instead of a blade angle). The 4-degree and 6-degree cases have a reduction of 40% in power at constant thrust. But the 8 and 10-degree cases have a reduction of 27% in power at constant thrust, and the 12 and 14-degree cases only have a 16% reduction in power.

Most of the above-cited literature is for helicopter applications, for understandable reasons. Vertical takeoff and landing UAVs (or "drones", or whatever is the currently fashionable moniker) are however likely to use propellers instead of untwisted rotors, likely of much smaller diameter and lower disk loading, than that which is expected for manned flight. There is also the intriguing question of the extent to which the direction of thrust matters – that is, whether the slipstream is into, or away from the ground plane. Finally, it would be appealing to find, without a too much poetic license, a collapse in the data for various propeller diameters, solidities, pitch-values and so forth, onto one curve of P/P_{OGE} vs. h/D.

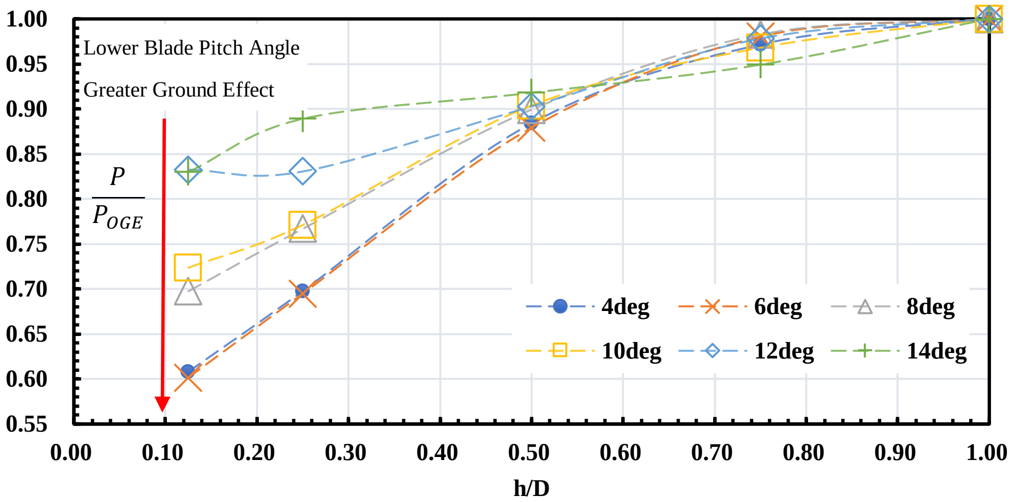

Figure 3. Knight and Heffner [5] data for normalized power ratio vs. h/D for various blade pitch angles.

II. Experimental Setup

Experiments were conducted on a thrust-stand at the University of Dayton. All propellers used in the experiment are from the "thin electric" APC series, selected because of evidence in the literature for their relatively high efficiency and good geometric repeatability across a broad range of diameters [9], [10]. All propellers were driven by an E-Flite Power 60-400kV brushless out-runner electric motor. Photographs are given in Figure 4, and a schematic in Figure 5. The motor is powered by a PSW 30-108 constant-voltage power supply and is attached to an ATI Industrial Automation Mini-40 six-component force balance (inside the plastic shroud shown in Figure 4; www.ati-ia.com [11]). Thrust is thus the axial-force along the balance axis of the sensor, and torque is a rotational moment about the balance axis of the Mini-40. The balance bolts to an aluminum-extrusion frame secured to the lab floor. The balance has a maximum of 120N force on its Z axis and maximum torque of 2 Nm about its Z axis. The encountered peak thrust and torque were 40N and 1.2 Nm, respectively. The data sampling rate was 100 Hz, for a 15-second duration at each data-point, with two runs to improve data repeatability for each data point (value of rpm, propeller diameter/pitch/type, and h/D value). In an effort to minimize sensor-drift, tare values are taken before and after each test, and then the average of the two tares is subtracted from the respective thrust and torque readings. The propeller orientation shown in Figure 4 is in "negative ground effect". That is, the pressure-side of the propeller is away from the ground plane.

With the thrust-line oriented horizontally, a vertical wall acts as the ground, and different h/D is achieved by sliding the thrust-stand's frame towards the wall. To attenuate blockage, the propeller-disk is >1 diameter behind any transverse members of the aluminum frame, and the 3D-printed plastic streamline-cowling covering the sensor is <3.3% of the propeller disk-area, for the case of the smallest propeller. As is usually the case, power-wires from the speed controller to the motor, cross the metric junction.

A photodiode is used to measure the propeller RPM. To achieve negative h/D, the propeller is attached reversely on the motor. The motor rotation direction is also reversed, achieved by exchanging two of the three wires between the motor and its electronic speed controller.

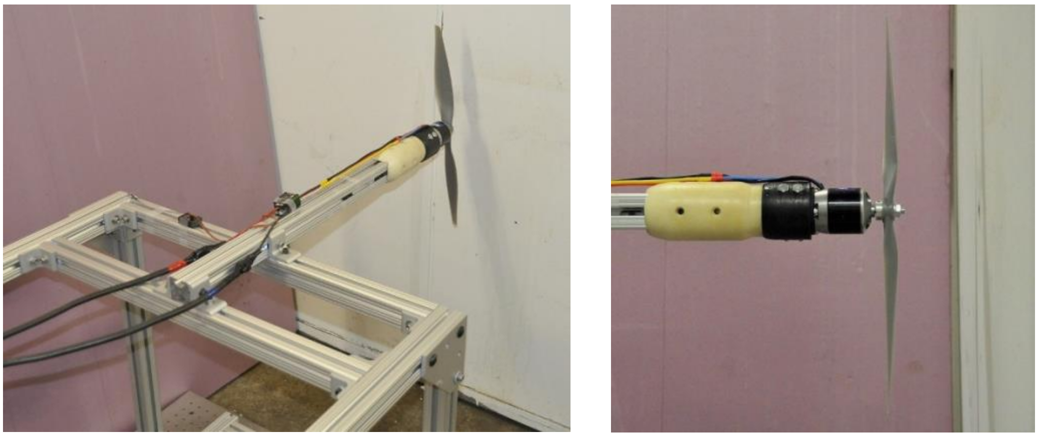

Figure 4. Photographs of thrust-stand (left) and propeller/motor/fairing (right).

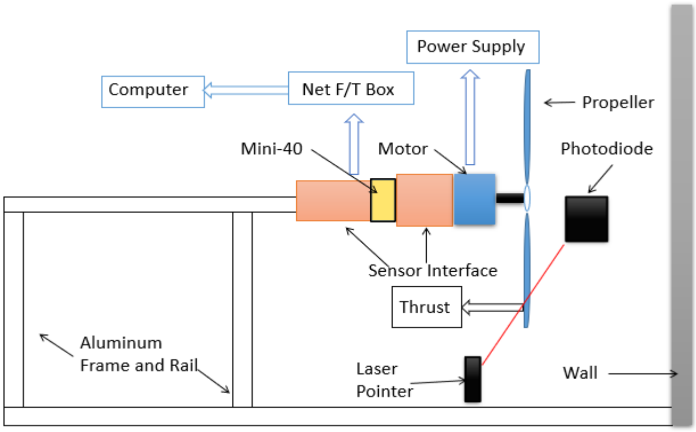

Figure 5. Schematic (side view) of thrust stand and ground-plane

Test parameters are shown in Table 1. The blade Reynolds number is calculated at 70% blade-span, for the peak rpm achieved for each given propeller.

Table 1. Propeller Properties

Propeller	Pitch (γ)/ Diameter (D)	Solidity (σ)	Peak Rotational Tip Speed (m/s)	Peak Reynolds Number (x1000)
17x12	0.705	0.0179	124.6	145
11x7	0.636	0.0225	100.1	100
11x7C	0.636	0.0340	96.3	155
17x10	0.588	0.0188	132.1	160
14x7	0.500	0.0193	122.3	130
14x7C	0.500	0.0270	119.9	190
11x5.5	0.500	0.0219	102.1	100
17x7	0.412	0.0211	142.0	190

The cases in Table 1 allow various parameter-combinations to elucidate trends. For 7” pitch, the three diameters (11”, 14” and 17”) comprise a study, oddly enough, of the effect of diameter – and less trivially, of the pitch to diameter ratio. Variation in solidity beyond what is available from the APC catalog was achieved by machining a larger propeller down to a smaller diameter. This preserves pitch. The resulting “cut” cases are labeled "C" in Table 1 and in the following graphs and discussions. All derive from cutting what was initially the 17x7" propeller. Several propellers observe a pitch-to-diameter ratio of 0.5, and in the case of the 17"-diameter propeller, a wide range of pitch-values is available. The rpm-range, with lower bound from signal-to-noise considerations, and upper bound limited by the power supply's capacity, was 3000-6000, depending on propeller diameter. The dimensionless ground-offset is -1 h/D to +1 h/D, with larger h/D deemed to be sufficiently close to infinity. For very small h/D, there is ambiguity as to where to prescribe the propeller center. For propellers with high blade twist and taper, at small nominal value of h/D, the actual h/D will vary considerably along the blade span. Here, we take as "zero" the midpoint of the propeller hub. For APC "thin electric" propellers, the hub is thicker (deeper), for larger values of blade pitch. This partially attenuates what would have been large proportional disparities in h/D between low-pitch and high-pitch propellers of the same diameter, for small h/D.

Before proceeding to IGE results, we show baseline OGE measurements with respect to existing comparisons. The present study's OGE results for the APC 11x7 propeller are compared with the UIUC database [12], and the earlier experiment by Gunasekaran et al. [13], also at the University of Dayton (Figure 6) but with a different experimental setup. The current data falls between the two others, and the difference of the data is less than 8% for both thrust coefficient and power coefficient. Interestingly, the two UD data sets almost overlap in thrust, within range of error-bars; meanwhile, the present UD set and the UIUC set have similar power values, also within error-bars. All three data sets have the same slope with respect to rpm: slightly positive for thrust, and essentially zero for power. The former has not yet been fully explained but could be a Reynolds-number effect.

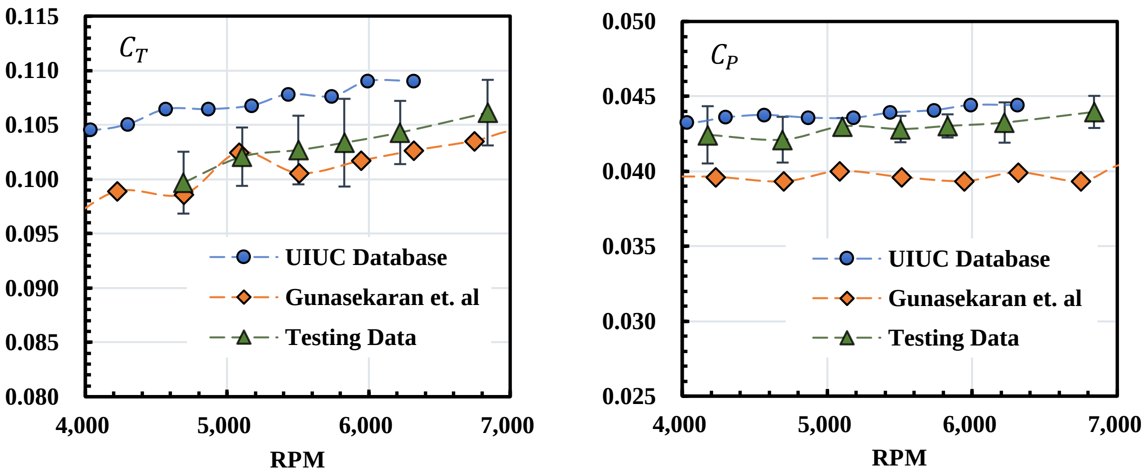

Figure 6. Comparison of OGE results for the APC 11x7 propeller, thrust (left) and power (right): present data, legacy University of Dayton data [12], and UIUC database [13]. Error-bars (95% confidence intervals) shown for current data set.

III. Results

A. Parameter Studies

We begin with a consideration of thrust coefficient and power coefficient vs. h/D and propeller rpm, for the 11x5.5 propeller (Figure 7). This is the smallest propeller in the present study, with the lowest operating Reynolds number (Table 1). There is a drift with increasing rpm, of a magnitude comparable to (and somewhat larger than) the error-bars for the data. The direction and magnitude of the drift in thrust coefficient is consistent with the thrust results of Figure 6. The drift in power coefficient is larger in Figure 7 than in Figure 6, but nevertheless, all rpm-curves follow the same trend. In the following, we average data across the measured rpm range, or interpolate, as need be. Also, data at h/D = 1 are almost indistinguishable from those at h/D = 1.5. This suggests that even |h/D| = 1 is approximate "infinity", or OGE. In the following, we limit the presentation to |h/D| ≤ 1, and normalize IGE results for the various coefficients, such that the IGE/OGE ratio at h/D = 1 is defined to be 1.

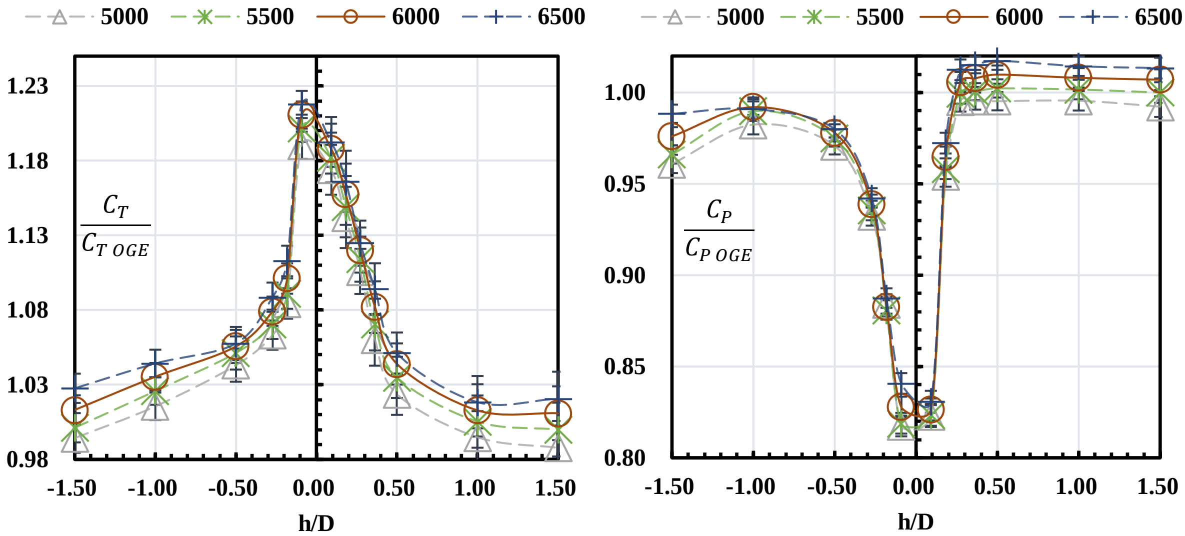

Figure 7. Ratio of IGE to OGE thrust (left) and power (right) for 11x5.5 propeller, at various test-rpm.

The alternative presentation is "normalized power", introduced above as P/P_{OGE}. To calculate this, a convenient (generally intermediate) value of rpm is fixed, and OGE thrust (that is, at h/D = 1) is calculated by interpolating across the available thrust data points across the rpm range. Then, for every h/D, one takes the OGE thrust value, and finds the IGE rpm at which thrust at that h/D equals the aforementioned OGE thrust. At the resulting rpm, one arrives at the power, *P*. Typical results for P/P_{OGE} are given in Figure 8.

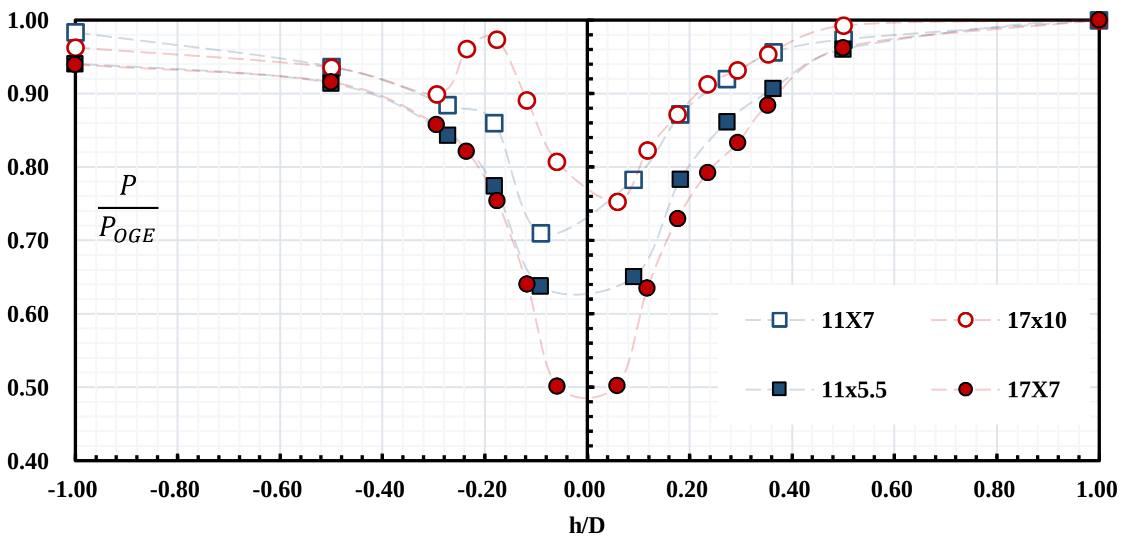

Figure 8. Normalized power required for constant thrust, for a selection of propellers.

For the 17x7 propeller, normalized power at the most extreme ground-separation is half of the OGE value. This is close to the prediction of Betz (Figure 1). But unlike the theory's prediction, the rise in P/P_{OGE} in stepping back to larger h/D is smooth, there is no reason to choose h/D = 0.3, or another other nearby value, as a demarcation point. The trend of P/P_{OGE} vs. h/D is symmetric, for small positive or negative h/D, but there is a slight asymmetry between h/D = + 1 and h/D = -1. For the 17x10 propeller, IGE power-reduction is much less dramatic, and is also asymmetric. For h/D ~ -0.2, there is an odd reversal in P/P_{OGE}. The two 11"-propellers are intermediate cases, with the 11x5.5 resembling more the 17x7, and the 11x7 resembling more the 17x10. Still, at least for positive h/D, the variation of P/P_{OGE} with h/D is smooth and orderly, resembling the experimental data in Figure 1.

P/P_{OGE} is a succinct presentation-format of IGE results, but it potentially masks to what extent thrust is increasing, or power-required is decreasing. A reversion to the more direct presentation of thrust and power coefficients is given in Figure 9. The trends of Figure 8 repeat in both subplots of Figure 9, with the exception that positive-negative h/D symmetry for the 17x7 case is not as evident. Instead, positive/negative trends in thrust and power combine, as it were, to produce the visually appealing symmetry for the 17x7 case in Figure 8. For both the 17x7 and 11x5.5, peak thrust-increase appears to be at slightly negative h/D, while peak power-reduction is at commensurately slightly positive h/D. The 17x10 case meanwhile still contains a curious reversal in both thrust and power at h/D ~ -0.2. For all propellers in Figure 8, reduction in power coefficient for the positive h/D range does appear to start near 0.3 h/D, consistent with Betz's proposed threshold.

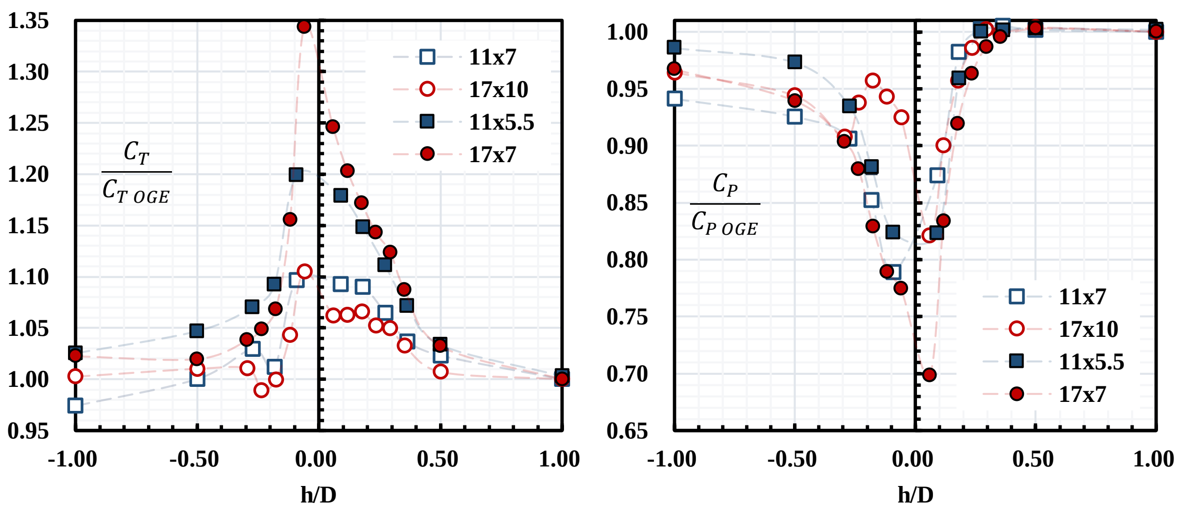

Figure 9. Ratio of IGE to OGE thrust coefficient (left) and power coefficient (right) for the cases of Figure 8

The variations for the four cases in Figure 8 and Figure 9 motivate the search for pitch-to-diameter ratio, or some other parameter, to arrive at a better collapse of the data. Figure 10 shows P/P_{OGE} for the full set of propellers, save for the cut-versions of the 17x7 propeller. In particular, there are three propellers of 7" pitch: 11", 14" and 17" diameter. Among these latter three, the trend is towards larger savings in IGE power, and better positive/negative h/D symmetry, with increasing diameter – which is to say, with decreasing ratio of pitch to diameter. The 11x5.5 and 14x7 have the same pitch-to-diameter (γ/D) ratio (= 0.5), and indeed, their P/P_{OGE} curves nearly overlap, the variation between the two being just short of being fully covered by their respective error-bars. However, the 11x7 has a higher pitch to diameter ratio than the 17x10, and yet it is the 17x10 which has the more anomalous P/P_{OGE} curve. To reiterate, in all cases, ground-proximity reduces P/P_{∞}, as expected; but the extent to which this happens, evidently depends on the pitch to diameter ratio.

Figure 11 replots the results of Figure 10, to emphasize pitch to diameter ratio, γ/D. For each h/D, P/P_{OGE} is seen to decline, as γ/D gets smaller. At the overlapping case, γ/D, P/P_{OGE} nearly overlaps for each h/D.

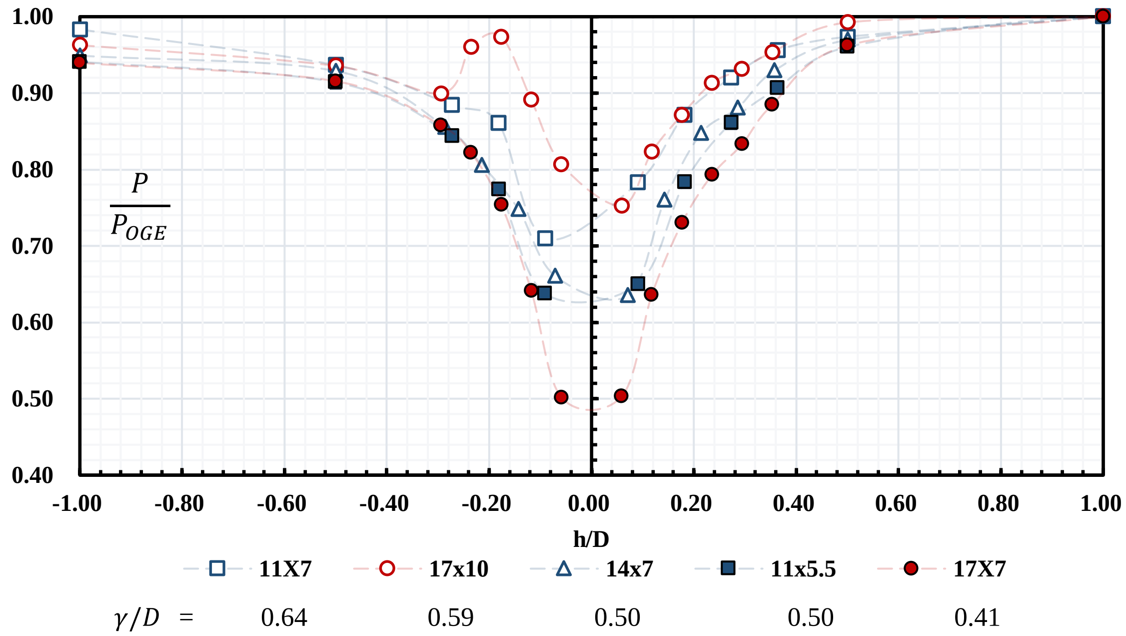

Figure 10. Normalized Power Required for Constant Thrust, for various propeller pitch to diameter ratios.

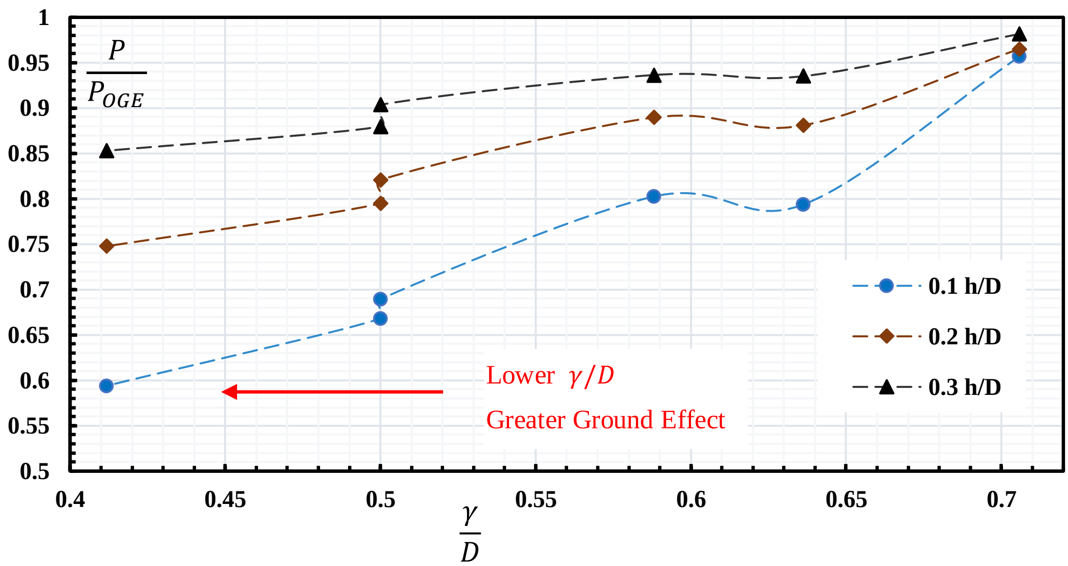

Figure 11. Normalized Power Required for Constant Thrust, with respect to propeller pitch to diameter ratio, at several different h/D values.

We next turn to a consideration of propeller solidity, σ. Figure 12 shows P/P_{OGE} vs. h/D, for 7”-pitch propellers: off-of-the-shelf APC 11x7 and 17x7, and cut-diameter versions of the 17” propeller (14” and 11” diameter). The 11x7 and 11x7C share γ/D (square symbols), as do the 14x7 and 14x7C (triangular symbols). For each such pair, the case with higher solidity has deeper P/P_{OGE} trough as h/D --> 0. Perhaps as mere coincidence, the 11x7C and 14x7 cases look to be nearly antisymmetric, despite their large disparity in γ/D (0.0340 and 0.0193, respectively).

Results of Figure 12 are replotted in Figure 13, terms of thrust and power coefficient vs. h/D. Once again, this reveals how usage of P/P_{OGE} is succinct, but can mask constituent trends. The two cut propellers (solid green symbols) clearly have larger IGE increase in thrust, but do not evince commensurate savings in power-coefficient. For all cases and both coefficients, scatter is higher for negative h/D than for positive. This observation is absent in Figure 12. Just as with pitch to diameter ratio, solidity is seen to matter – but a definitive relationship remains elusive.

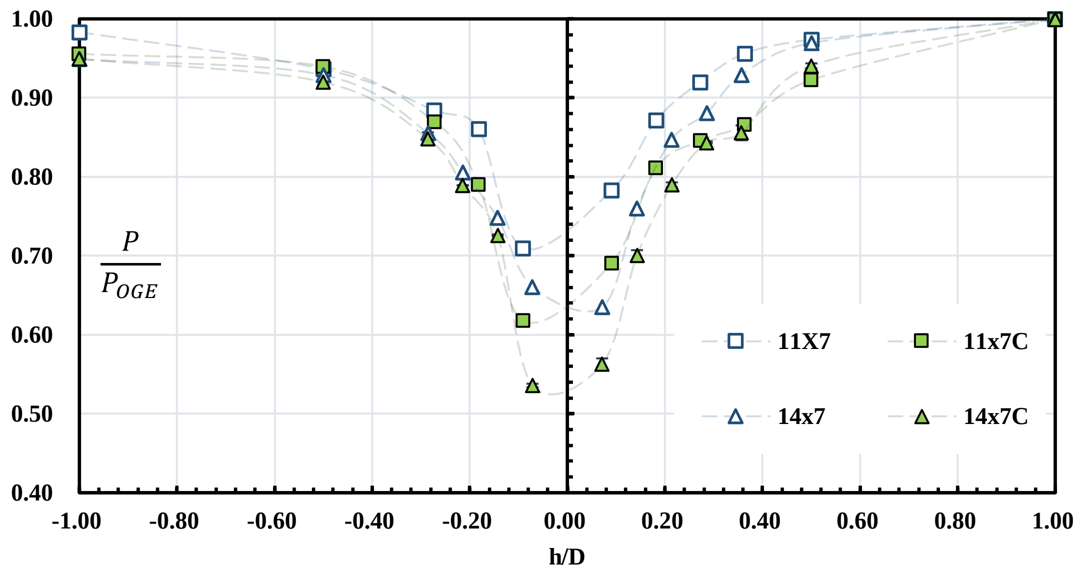

Figure 12. Normalized Power Required for Constant Thrust, for cases of higher or lower solidity.

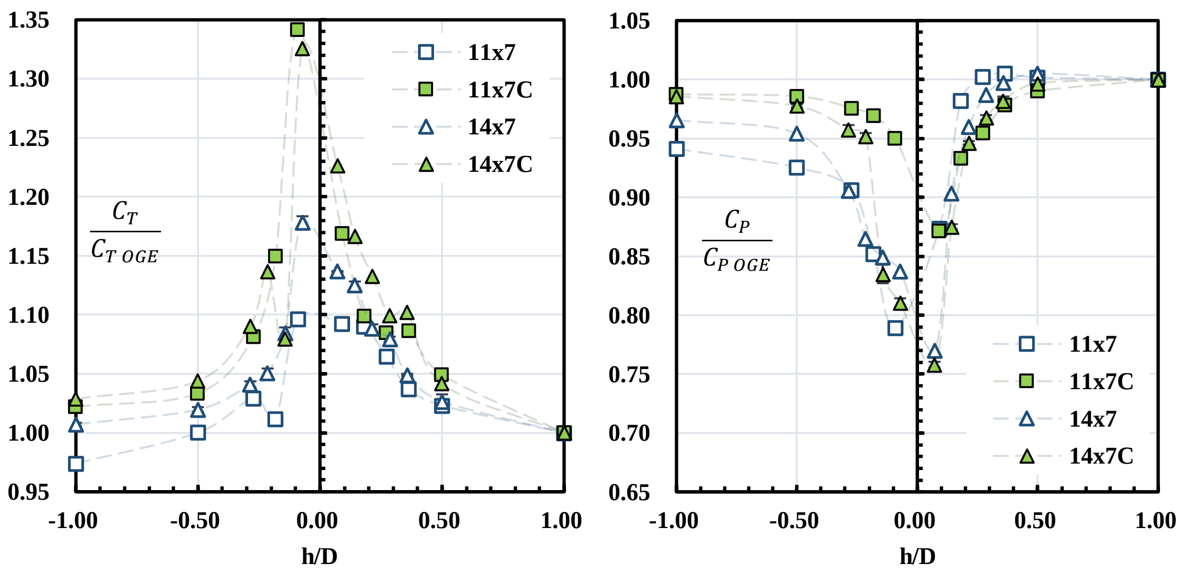

Figure 13. Normalized thrust coefficient (left) and power coefficient (right) from the results of Figure 12

B. Synthesis

With the suggestive but not definitive role of pitch-to-diameter-ratio and solidity, in IGE thrust or power vs. h/D, it is appealing to consider an empirical relationship that might collapse the data to one curve. While a mathematical optimization routine might efficiently do this, we pursued manual methods. The result is Eqn. 1:

$$\frac{P_c}{P_{OGE}} = 1 - \frac{1 - \frac{P}{P_{OGE}}}{\sigma * 100}\left(\frac{P_a}{D}\right)^n \tag{1}$$

Where n = 2 at positive h/D range, and n = 1 at negative h/D range.

Figure 14 shows the power required at constant thrust, corrected by Equation 1. For positive h/D, curves for most propellers nearly collapse. For the negative h/D, collapse is somewhat worse than for positive h/D. The 17x10 is a partial outlier, with its anomalous trend in P/P_{OGE} at h/D ~ -0.2 is not resolved by equation 1. More troublesome is the 17x12 propeller, which evinces very little IGE benefit. It only loosely follows equation 1 for positive h/D, and is completely off for negative h/D. It has the highest γ/D ratio in this study. One might speculate that poor IGE results are due to blade stall, and indeed, the onset of stall may account for why the higher pitch-to-diameter propellers as a group do worse than those with lower γ/D. That is, if ground-effect increases the effective angle of attack, it may exacerbate a tendency towards stall. But in extreme ground effect, exactly what happens to the flow field is unclear. A flow visualization campaign may settle the matter.

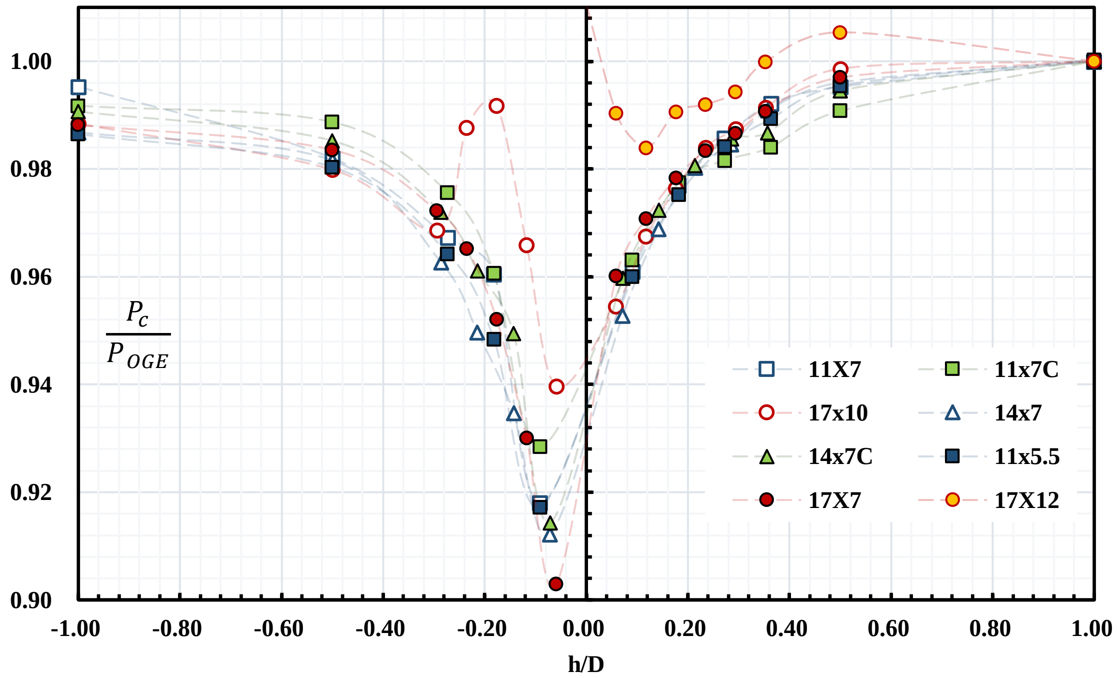

Figure 14. Correction of Normalized Power Required at Constant Thrust

IV. Conclusion

Ground effect was found, not surprisingly, to increase effective thrust, and to decrease power-required, for a given propeller rotational speed. Varying propeller solidity and pitch-to-diameter ratio show that, in general, higher solidity and lower pitch-to-diameter ratio both lead to stronger ground effect. For the highest ratio of pitch to diameter ratio, there are anomalous results in extreme ground effect, possibly due to blade stall. For the lowest ratios of the pitch to diameter ratio, Betz's theory is matched as h/D --> 0 reasonably well, perhaps by coincidence. For larger pitch-do-diameter ratios, the theory overpredicts ground-effect. Combination of the various parameters leads to a phenomenological relation, according to which the curve of normalized power at constant thrust, vs. ratio of ground separation to diameter, collapses fairly well, for all propellers save for those of the largest pitch-to-diameter ratio. The relation holds for positive or negative ground effect; that is, whether the pressure-side of the propeller is away from, or towards the ground, respectively. Propellers of higher solidity than that available off of the shelf, were made by cutting the diameter of larger propellers; this also preserves pitch. These were also shown to follow the phenomenological relation. A broader parameter study, to see if the phenomenological model still holds, could be exercised by combining several off-of-the-shelf propellers into one of the multiple blades, which of course entails some mechanical difficulty. Further insight would also be gleaned from visualization of the flow field, to ascertain if (and how) higher pitch-to-diameter propellers may stall in extreme ground effect, and if such stall is particularly acute along those portions of the blade which, owing to the propeller's geometry, come closest to the ground plane.

V. References

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